

Signaling Helmet

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1 Introduction

Cycling is an effective and environmentally friendly method of transportation that motivates environment-friendly behavior and sustainable and rewarding lifestyle habits (Pucher & Buehler, 2017). However, cyclist's safety remains a concern, as bicycles are missing built-in turn signals, which makes communication with other road users difficult. Cyclists usually use hand signals, which require them to take one hand off the handlebar, this worsens their control on the bike and increases the risk of accidents, especially in low visibility, darkness, or heavy traffic. The traditional safety of cyclist focused mainly on wearing the helmet. Many studies show that preventing strategies are needed to reduce the accidents which many single bicycle crashes led to. Therefore, other protection strategies are needed as well (Ohlin et al., 2019). To help with this problem, we created a signaling helmet with LED indicators and gesture-controlled gloves. The system allows cyclists to activate turn signals with a simple finger gesture, without removing their hands from the handlebars. By improving visibility and communication between cyclists, pedestrians, and other road users, this solution aims to enhance cyclist visibility and reduce traffic accidents. This report discusses the design process from ideation to the final prototype. Additionally, it suggests possible improvements for future development.

2 Ideation

Our group of 5 members aimed to create a wearable system that could make difference in the real life world. Therefore, we have chosen the theme wearable future product and moved to the brainstorming session.

2.1 Brainstorming and Sketching

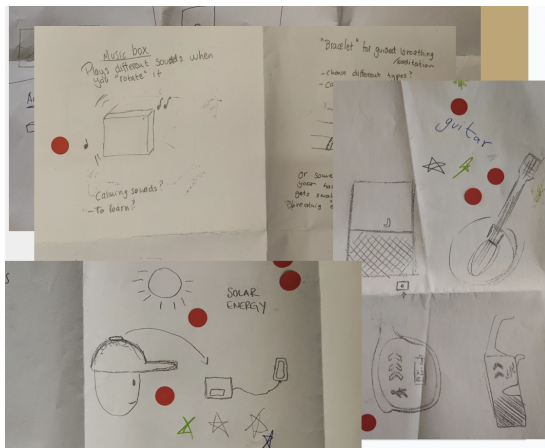
At the beginning, our group researched tangible products online across various themes to find inspiration and see what other designers had created. During the brainstorming session, we sketched our ideas and wrote down our notes and thoughts to share with each other. Each team member found at least three ideas and shared them with the group. The choice to brainstorm was an effective method for enhancing a group's creativity by helping them generate, develop, and share ideas to find practical solutions (Al-Samarraie & Hurmuzan, 2018).

Furthermore, we narrowed down our ideas by voting and selecting four key concepts. We used the dot voting method (Dam & Siang, 2018), where each team member placed two pieces of tape on their favorite concepts. The ideas with the most votes were chosen, see figure 1a for reference. The selected concepts were:

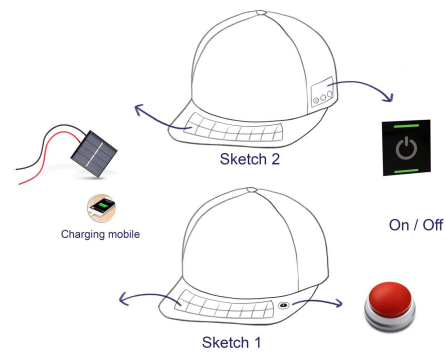
- A solar energy hat as a wearable which use electrical energy for a power bank, fan, and headphones.
- A guitar, which teach people to play music using light and at the same time it can hear a tone and show how it can be played by the user.

- Garbage bin that play happy music when the user cast right material into the right bin and sad music otherwise; to help users to have a better garbage waste management.
- Two music boxes that two users have in their homes, these devices are connected to each other. Using them, the users can interact with each other by sending emotional messages as for example warm light or calm sound to the other user.

Later, we used the Six Thinking Hats method to discuss and analyze these four ideas from multiple perspectives (Vedawala et al., 2024). This helped us identify the advantages and disadvantages of each idea, allowing us to make a better decision. After this, the dot voting method to choose the final idea was used again. The final choice landed on the solar energy wearable, designed in the form of a hat. Apart of the solar energy sensor, the hat will include a power bank and tactile buttons that activates features like a fan or speaker as shown in figure 1b.



(a) Our initial ideas



(b) The chosen initial idea

Figure 1: Brainstorming and sketching

2.2 Project Idea Modification

We presented our idea during the project pitch, we received feedback on the solar sensor's ability to charge batteries and realized that it does not give enough power to the battery to active all of the features we planned to have. Such a small area as on top of a hat only allows for small solar panels. Additionally, Sweden's weather does not provide enough sunlight every day comparing with another warmer countries.

After deep consideration of our project idea, we decided to modify it while remaining within the theme of wearable. We have chosen to focus on solving a critical problem faced by cyclists, which affects their safety as well as the safety of pedestrians and other road users who may be involved in accidents with bicycles. The number of death in traffic accidents in Europe is decreasing significantly but, unfortunately, the number of death between cyclists does not follow the same trend and cyclists death become a larger number every year (Uittenbogaard et al., 2020) therefore, we have a challenge to create a product that can enhance the safety and improve the communication and the visibility among the cyclists and other roads users in form of wearable.

3 Problem Statement and Solution

The problem lies in the difficulty of using effective turn signals, as most bicycles, especially older ones, they don't have built-in lights that function as turn indicators. As a result, cyclists use arm signals to alert others on the road. However, this requires raising a hand from the handlebars, as illustrated in figure 2, which reduces control of the bicycle and increases risk of accidents. (Johnsen & Funk, 2019). It is also

known specially the older cyclists find it difficult to releasing the handlebar during biking for gesturing and turning the head to look behind before turning (Bernhoft & Carstensen, 2008).



Figure 2: Cyclists signal before turning.

Swedish cyclists face a significantly higher risk compared to car occupants, with a 10 times greater chance of fatal accidents and a 20 times higher likelihood of injuries requiring hospital treatment for every million kilometers traveled (Nilsson et al., 2017).

We decided to develop a helmet equipped with LED strips turn indicators on its sides. These indicators will be activated by gestures using the gloves. The aim of this system to enhance communication and improve visibility between cyclists and other road users. The gloves are equipped with flex sensors attached to the index fingers. These sensors will send a signal to the LED strip in the helmet when they are bent and the bending happens when the gesture is made. This design allows cyclists to signal their direction before turning without removing their hands from the handlebars, thus enhancing safety while riding.

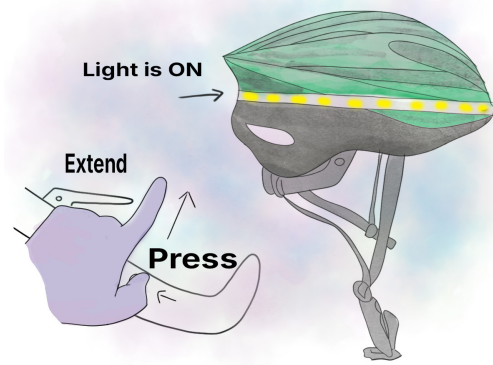
4 Prototyping

Before constructing the prototype, we conducted interviews with two cyclists to understand how they naturally move their fingers while riding. This research highlighted the importance of preventing accidental activation of the turn signal on the helmet when cyclists move their fingers unintentionally. To address this, we added a pressure sensor into the glove, requiring cyclists to press their thumb while raising their index finger to active the LED indicator, as illustrated in Figure 3a. We have chosen using the thumb and the index finger because they have shown to be intuitive gestures (Xiao & He, 2019). This design minimizes the risk of unintended activation.

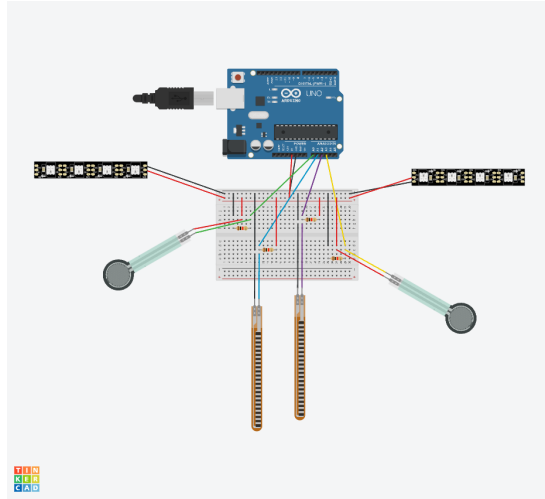
Our initial prototype was constructed using an Arduino UNO, a breadboard, wires, resistors, two flex sensors, two pressure sensors, and two LED strips. Additionally, we created a digital version of the prototype using Tinkercad, as shown in Figure 3b.

Before working on the final prototype, we first tested the sensors on a breadboard to measure resistance values and determine the appropriate activation threshold. Once the sensors functioned as expected, we soldered the wires to the two LED strips and attached them to the helmet.

Initially, the sensors were sewn into the gloves using conductive thread, but this resulted in unstable connections. To improve reliability, we soldered and glued the wires directly to the sensors and secured them to the exterior of the gloves, ensuring better stability. Additionally, we adjusted the threshold values multiple times to ensure the LED stripes activation was precise without being overly sensitive.



(a) Sketch of the signaling helmet



(b) The prototype build by Tinkercad

Figure 3: The prototype

5 Evaluation

We conducted a two-week evaluation to gather user feedback. The first round focused on cyclists testing the prototype, while the second round involved pedestrians assessing how effectively the signaling system communicated the cyclist's intentions. The user testing aimed to answer the question: How can a wearable system, consisting of a helmet and gloves, enhance communication between cyclists and other road users? Additionally, we explored the following key questions:

- How visible is it that the cyclist is going to turn?
- Is the gesturing intuitive and easy to do while cycling?
- Does the LED color make sense for the purpose?
- Is the light visible in different conditions?

We tested the prototype with six users of different ages and genders, including teachers, students, and regular cyclists. Participants were recruited through emails, direct invitations, and by approaching cyclists in public. Each test session lasted about 10 minutes, during which we observed user behavior, gathered feedback, and asked participants to fill out a questionnaire to document their experiences.

Cyclists mentioned that they couldn't see the LED on their helmet, so they suggested adding some form of feedback to confirm activation. Some participants suggested changing the LED colors, adjusting the blinking patterns, and increasing the number of LEDs for better visibility. Many cyclists also expressed the need for a way to confirm whether their signal was active, such as a vibration or sound notification. Users had mixed opinions on whether the LED should turn off automatically, require manual deactivation, or offer a customization option.

6 Improvements

After considering the feedback, we made several changes to improve the prototype:

1. The LED color changed. Some users pointed out that the original blue LED referred to the emergency vehicle signals, such as police lights. To avoid confusion, we switched the LED to orange, making them more like traditional turn signals in cars.

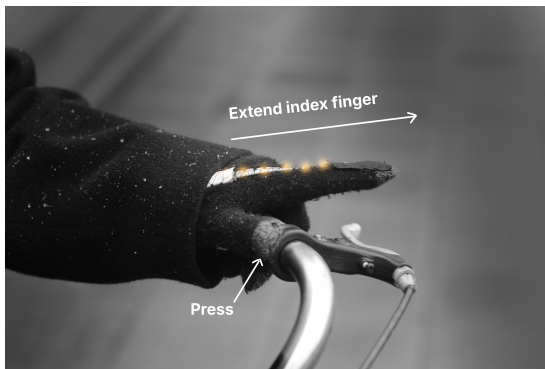
2. We did adjustment to the light patterns. Originally, the LED strips blinked in groups of three, moving from back to front. To improve visibility, we modified the pattern so that all LED blink at the same time which.

As a future improvement, we also plan to add a forward facing flashlight on the helmet, providing cyclists with better visibility of the road ahead in dark trips.

7 Limitations

Based on user behavior, we realized the need to add a vibration or a synchronized light on the gloves, allowing cyclists to confirm when the signal was active while gesturing, as illustrated in Figure 4. However, due to the limited time available for this project, we decided to leave this feature for future development.

Additionally, the ability for users to customize the light color or brightness was missing from our prototype. Another key issue was the wired connection between the gloves and the helmet, which is unnecessary. A wireless solution, such as Bluetooth, would be a better alternative. Moreover, wearing gloves in hot weather is uncomfortable, making the product less desirable for some users.



(a) Gesturing feedback



(b) The signal helmet and gesturing feedback

Figure 4: Visual feedback on glove during signaling

8 Conclusion

Our project aimed to improve cyclist safety by developing a wearable turn signaling system using a helmet and gloves equipped with gesture-activated sensors. This convert made the project more relevant and meaningful, while it took a real-world safety issue rather than just focusing on sustainability. User testing and feedback highlighted areas for further improvement. The interactions with other road users are an important factor for safety (Westerhuis et al., 2021), By improving communication between cyclists, pedestrians, and other road users, the product has the possibility to reduce cycling-related accidents and contribute to have safer streets for everyone.

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